5.4 Solving Radical Equations and Inequalities

Name _____ Period ____ Date _____

Algebra II 5.4 · Student sheet

I can…

- solve equations that contain radicals or rational exponents.
- identify extraneous solutions.
- solve radical inequalities.

Vocabulary · fill in the missing word

Radical equation — an equation with a _____ in a radicand

Extraneous solution — an _____ solution that does not work in the original equation

Radical inequality — an inequality with a _____ in a radicand; the radicand must also be ≥ 0 for even roots

Part 1 · Find the impostor

Solve $\sqrt{(x + 2)} = x$ by squaring both sides. Show your work.

Answer	Check it in $\sqrt{(x + 2)} = x$	Does it work?
x =		
x =		

Graph $y = \sqrt{(x + 2)}$ and $y = x$ in Desmos. How many times do they cross? _____

Why did one answer fail?

Part 2 · Notes and examples (fill in each note, then work the example)

Solving a radical equation: Step 1: _____ the radical. Step 2: raise both sides to the same _____ as the index. Step 3: solve. Step 4: _____ every answer in the _____ equation.

Example 1

Solve $2\sqrt{x} + 5 = 11$.

Example 2

Solve $\sqrt[3]{(2x - 9)} - 1 = 2$.

Extraneous solution: An **extraneous solution** is an answer that _____.

Example 3

Solve $x + 1 = \sqrt{7x + 15}$.

Example 4

Solve $\sqrt{2x + 3} = \sqrt{x + 7}$.

Rational exponents: To solve x^(m/n) = b, raise both sides to the _____ power.

Example 5

Solve $x^{(\frac{3}{2})} = 8$.

Radical inequalities: For an inequality with an even root, the radicand must also be _____.

Example 6

Solve $3\sqrt{x - 1} \leq 12$.

Part 3 · Partner practice

Partner 1

Solve $\sqrt{4x - 3} + 2 = 7$.

Partner 2

Solve $x - 6 = \sqrt{3x}$.